

Project Implementation Plan

1. Team composition

No.	Name	Group	Role
1.	Boga Karla-Maria	30333/1	Team Leader
2.	Gota Timeea	30333/2	Back-End Developer
3.	Toader Diana-Patricia	30333/2	Fullstack Developer
4.	Runcan Maria	30133/2	Front-end Developer
5.	Gherasim Paula-Alexandra	30133/2	Tester

2. Project Topic

Title: **Restaurant Management System**

Description:

The Restaurant Management System is a full-stack software solution designed to streamline and automate the daily operations of a restaurant. It integrates order processing, inventory management, staff coordination, billing, and analytics into a single platform, improving efficiency, accuracy, and customer experience.

The system supports multiple user roles such as administrators, waitstaff and kitchen staff, ensuring smooth communication between front-of-house and back-of-house operations.

Core Features:

- **Order Management**
 - Digital menu with categories (appetizers, main course, drinks, desserts)
 - Table-based order assignment
 - Modify/cancel orders
- **Billing & Payments**
 - Automatic bill generation
 - Discount calculation
 - Multiple payment methods (cash, card)
- **Kitchen Management**
 - Kitchen Display System

- Status updates: Pending, In progress, Ready to serve
- Estimated preparation time tracking

- **Inventory Management**

- Track raw materials and ingredients
- Low-stock alerts and notifications
- Supplier management and restocking logs

- **Staff Management**

- Role-based access control (Admin, Waiter, Chef)
- Shift scheduling

- **Reporting & Analytics**

- Best-selling items analysis
- Revenue breakdown by category
- Inventory usage reports

- **Notifications & Alerts**

- Order ready alerts for waitstaff
- Low inventory notifications
- System alerts for admins

3. The implementation plan

Week 6-7 : Software Modeling and Design Specifications

Week 8-9 : Implementation of the application

Week 10-11: Fixing existing bugs

Weeks 12-14 : Implementation of new features

4. Task list

General Tasks

- Define the project scope and system requirements
- Select technologies (C#, .NET, SQL Server)
- Establish application architecture (UI, Business Logic, Data Access layers)
- Set up version control (Git) and workflow
- Organize regular team meetings and track progress
- Prepare project documentation and reports

Team Leader Responsibilities (Boga Karla-Maria)

- Coordinate team activities and ensure effective communication
- Assign tasks and establish internal deadlines
- Monitor team progress and evaluate contributions
- Design the overall system architecture
- Contribute to UML diagrams and class design
- Define application structure and modules
- Assist in GUI design and data persistence decisions
- Review deliverables and request improvements if needed
- Ensure all requirements are met
- Prepare and maintain documentation (Implementation Plan, Design Specifications, Final Report)

Back-End Developer Responsibilities (Gota Timeea)

- Design and implement the database schema
- Develop core business logic
- Implement data access layer
- Create and manage classes for:
 - Orders

- Users and roles
 - Inventory
 - Billing
- Handle data validation and security
- Ensure proper communication between application layers
- Optimize performance and data handling

Fullstack Developer Responsibilities (Toader Diana-Patricia)

- Integrate front-end and back-end components
- Implement complete functionalities across layers
- Ensure smooth data flow between UI and database
- Contribute to both UI and backend logic
- Handle errors and edge cases
- Support performance optimization

Front-End Developer Responsibilities (Runcan Maria)

- Design and implement the graphical user interface
- Create application screens:
 - Dashboard
 - Menu and ordering interface
 - Administration panel
- Ensure responsive and user-friendly design
- Validate user input and improve usability
- Collaborate with backend for data integration

Tester Responsibilities (Gherasim Paula-Alexandra)

- Define test cases before implementation
- Create unit and functional tests
- Test application features and workflows

- Identify and report bugs
- Perform regression and integration testing
- Validate that requirements are correctly implemented

Functional Tasks

Order Management

- Implement digital menu with categories
- Create table-based order system
- Allow order modification and cancellation

Billing and Payments

- Implement automatic bill generation
- Add discount calculation
- Support multiple payment methods

Kitchen Management

- Implement order status tracking (Pending, In Progress, Ready)
- Create kitchen display interface
- Estimate preparation times

Inventory Management

- Track ingredients and stock levels
- Implement low-stock alerts
- Manage suppliers and restocking

Staff Management

- Implement role-based access control (Admin, Waiter, Chef)
- Manage staff scheduling

Reporting and Analytics

- Generate sales reports
- Identify best-selling items

- Analyze revenue by category
- Track inventory usage

Notifications and Alerts

- Notify staff when orders are ready
- Alert for low inventory
- Provide system notifications for administrators